

AA/EE/ME 548

Linear Multivariable Control
Control Certificate Functions

Spring 2026

Recall: Stability

Let x_{eq} be an equilibrium point ($f(x_{eq}) = 0$)

- **Lyapunov stability:** If start near x_{eq} , will remain near x_{eq}
- **Asymptotic stability:** If start near x_{eq} , will eventually converge to x_{eq}
- **Exponential stability:** If start near x_{eq} , will converge to x_{eq} at exponential rate

Local vs global stability

Recall: Lyapunov function

Consider $\dot{x} = f(x)$ where $x \in \mathcal{X}$. Without loss of generality, let $x_{eq} = 0 \in \mathcal{X}$. Let $V: \mathbb{R}^n \rightarrow \mathbb{R}$ be a scalar, continuously differentiable function (a Lyapunov candidate). Then the equilibrium $x_{eq} = 0$ is Lyapunov stable if:

- $V(0) = 0$
- $V(x) > 0 \ \forall x \in \mathcal{X} \setminus \{0\}$
- $\dot{V}(x) = \nabla V(x)^T f(x) \leq 0 \ \forall x \in \mathcal{X}$

Level sets of Lyapunov Function are forward invariant sets

Consider a dynamical system defined by the differential equation

$$\dot{x} = f(x)$$

A set $\mathcal{C}$ is **forward invariant** with respect to this system if:

$$x(0) \in \mathcal{C} \Rightarrow x(t) \in \mathcal{C} \quad \forall t \geq 0$$

If we define the set using a continuously differentiable function $h(x)$, such that

$$\mathcal{C} = \{x \in \mathbb{R}^n : h(x) \leq 0\}$$

Control Lyapunov Function

What if we had $\dot{x} = f(x, u)$?

Even if $\dot{x} = f(x, 0)$ is unstable, we could choose u to ensure the system remains stable.

Control Lyapunov function

Consider $\dot{x} = f(x, u)$ where $x \in \mathcal{X}$, $u \in \mathcal{U}$. Without loss of generality, let $x_{eq} = 0 \in \mathcal{X}$. Let $V: \mathbb{R}^n \rightarrow \mathbb{R}$ be a scalar, continuously differentiable function (a Lyapunov candidate). Then the equilibrium $x_{eq} = 0$ is Lyapunov stable if:

- $V(0) = 0$
- $V(x) > 0 \ \forall x \in \mathcal{X} \setminus \{0\}$
- $\min_{u \in \mathcal{U}} \nabla V(x)^T f(x, u) \leq 0 \ \forall x \in \mathcal{X}$

Level sets of Control Lyapunov Function are **control invariant sets**

Consider a dynamical system
defined by the differential equation

$$\dot{x} = f(x, u)$$

A set $\mathcal{C}$ is **control invariant** with
respect to this system if:

$$\exists u(\cdot) \in \mathbb{U} \text{ s.t. } x(0) \in \mathcal{C} \Rightarrow x(t) \in \mathcal{C} \quad \forall t \geq 0$$

What if you want to stay inside a set,
rather than converge to a point?

Barrier Functions

Consider $\dot{x} = f(x)$. Define $\mathcal{S} = \{x \in \mathcal{X} \mid b(x) \geq 0\}$ where $b: \mathbb{R}^n \rightarrow \mathbb{R}$. If $x(0) \in \mathcal{S}$ and $\nabla b(x)^T f(x) \geq 0$ when $b(x) = 0$, then $b(\cdot)$ is a **barrier function**.

$\Rightarrow \mathcal{S}$ is forward invariant.

Control Barrier Functions

Consider $\dot{x} = f(x, u)$. Define $\mathcal{S} = \{x \in \mathcal{X} \mid b(x) \geq 0\}$ where $b: \mathbb{R}^n \rightarrow \mathbb{R}$. If for all x where $b(x) = 0$, we have

$$\max_{u \in \mathcal{U}} \nabla b(x)^T f(x, u) \geq 0$$

Then $\mathcal{S}$ is control invariant.

Control Barrier Functions – beyond the boundary

In most practical scenarios, we prefer a more progressive constraint as the state approaches the boundary – the closer you get to the boundary, the tighter the restriction on its motion, farther away, there should be more flexibility.

(Extended) Class $\mathcal{K}$ Functions

Class $\mathcal{K}$ and Extended Class $\mathcal{K}$ Functions

A Lipschitz continuous function $\alpha : [0, a) \rightarrow [0, \infty)$, $a > 0$, is said to belong to class $\mathcal{K}$ if it is strictly increasing and $\alpha(0) = 0$. Moreover, $\alpha : [-b, a) \rightarrow [-\infty, \infty)$, $a > 0$, $b > 0$, is said to belong to the *extended* class $\mathcal{K}$ if it is strictly increasing and $\alpha(0) = 0$.

Control Barrier Functions

Consider $\dot{x} = f(x, u)$. Define $\mathcal{S} = \{x \in \mathcal{X} \mid b(x) \geq 0\}$ where $b: \mathbb{R}^n \rightarrow \mathbb{R}$. The function $b(\cdot)$ is a **control barrier function** for the system if there exists a(n) (extended) class $\mathcal{K}$ function α such that

$$\max_{u \in \mathcal{U}} \nabla b(x)^T f(x, u) \geq -\alpha(b(x)) \quad \forall x \in \mathcal{X}$$

$\Rightarrow \mathcal{S}$ is control invariant.

Stabilizing and safe control sets

CLF: Stabilizing controls:

$$\mathcal{U}_{\text{stab}}(x) = \{ u \in \mathcal{U} \mid \nabla V(x)^T f(x, u) \leq 0 \}$$

CBF: Safety-preserving controls:

$$\mathcal{U}_{\text{safe}}(x) = \{ u \in \mathcal{U} \mid \nabla b(x)^T f(x, u) \geq -\alpha(b(x)) \}$$

Stabilizing and safety filters

We can use CLFs and CBFs to ensure stability and safety properties are “satisfied.”

Given current state x and desired control u_{des} :

Why are control affine dynamics useful?

$$\mathcal{U}_{\text{stab}}(x) = \{ u \in \mathcal{U} \mid \nabla V(x)^T f(x, u) \leq 0 \}$$

$$\mathcal{U}_{\text{safe}}(x) = \{ u \in \mathcal{U} \mid \nabla b(x)^T f(x, u) \geq -\alpha(b(x)) \}$$

CBF-CLF-QP filter

CBF-CLF-QP filter

Further thoughts (discuss with neighbor)

- Obtaining a valid CLF/CBF
- Relative degree of system
 - What if $\nabla b(x)^T B(x) = 0$ for all $x \in \mathcal{X}$?
- Balance between safety/stability and performance (graceful degradation)
- Discrete-time setting?
- Requirements on the functions and dynamics?

Further reading and videos

- [High-order CBFs.](#)
- Survey paper: [Neural CBFs and CLFs](#)
- [Learning CBFs from demonstrations](#)
- Survey paper from IEEE Control Systems Magazine on [Data-Driven Safety Filters](#)
- [Professor Aaron Ames \(Caltech\)](#) on CBFs: [Recorded talk](#)
- [Professor Sylvia Herbert \(UCSD\)](#) on connecting CBFs with HJ reachability (aka optimal control): [Recorded talk](#)
- Learning reasonable human behaviors for safe set computation: [Paper](#)
- Multi-agent interaction learning with CBFs (guest speaker later): [Paper](#)